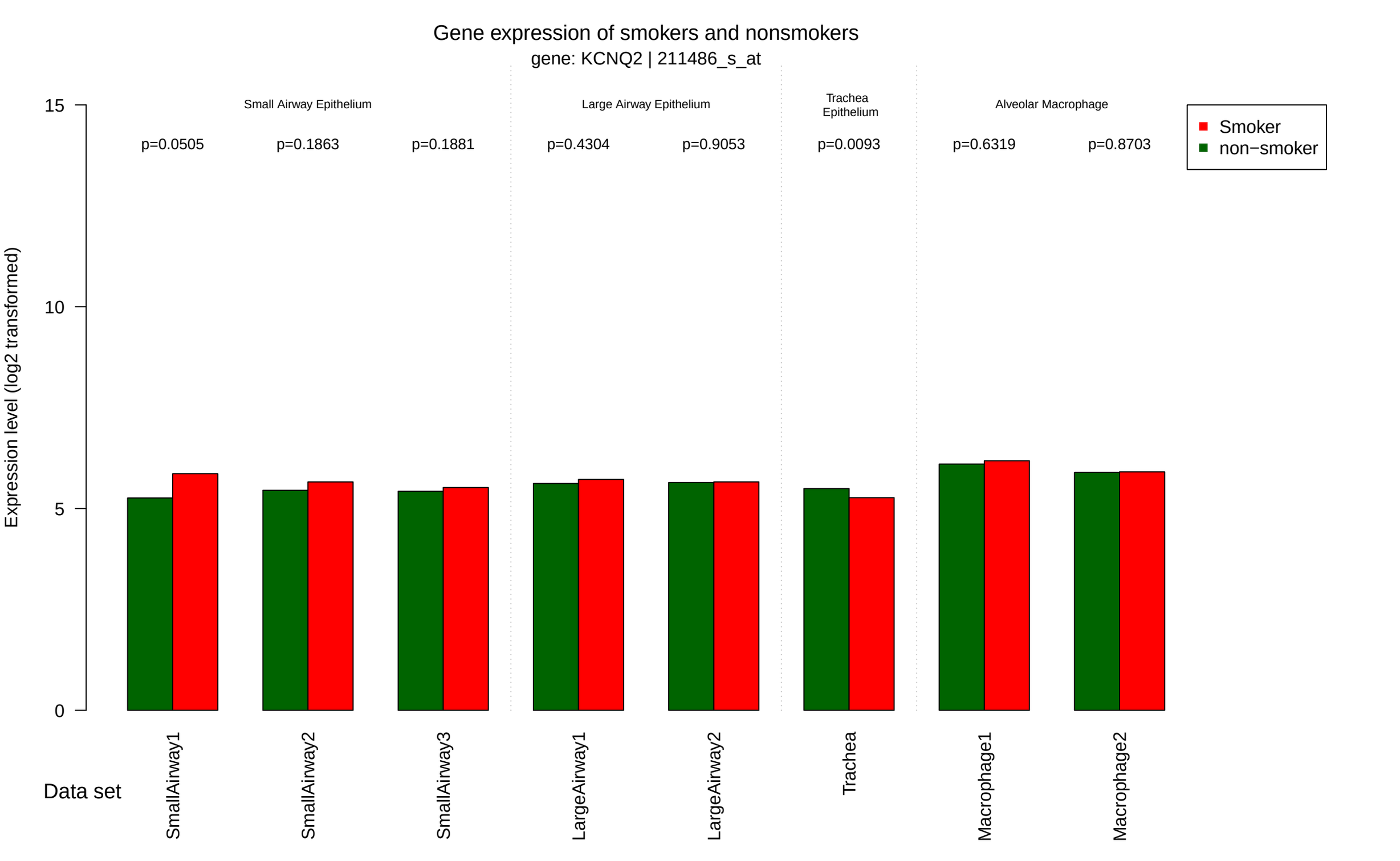
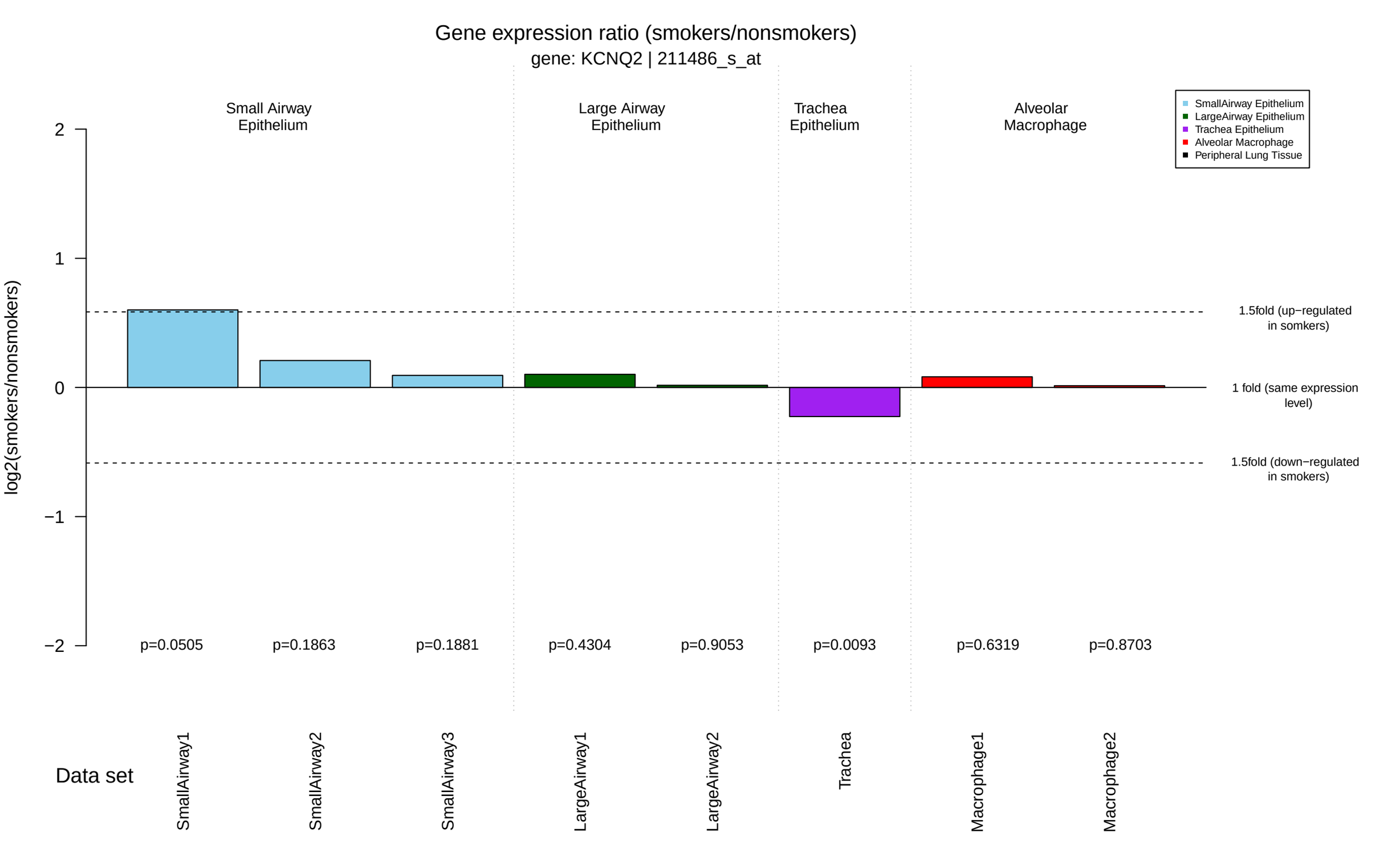
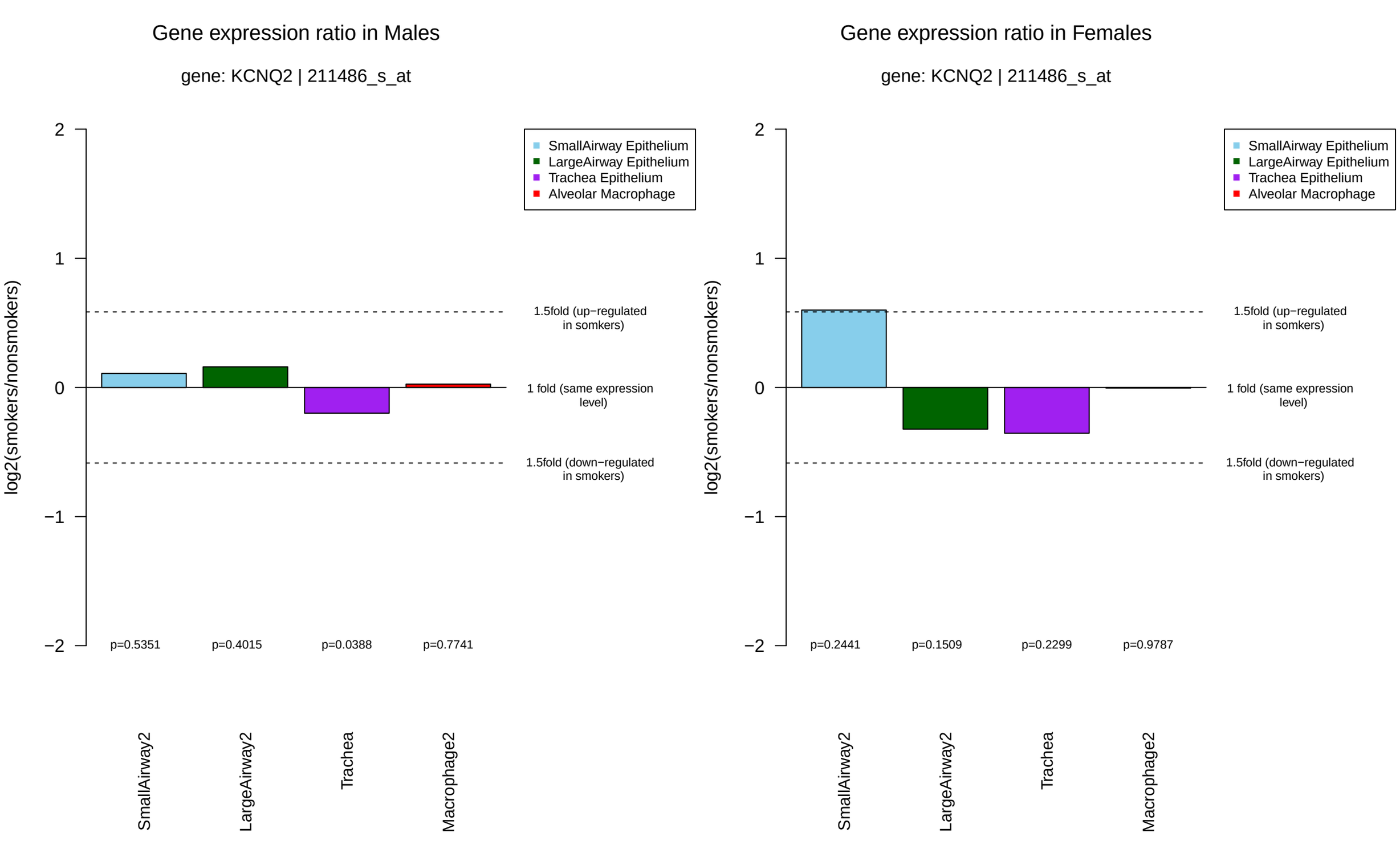
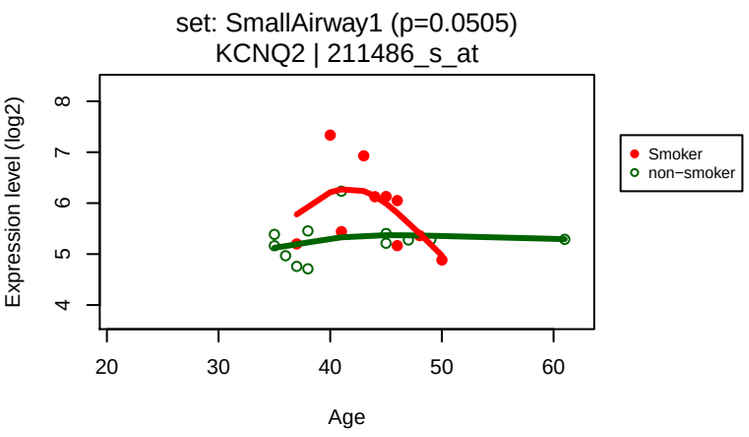
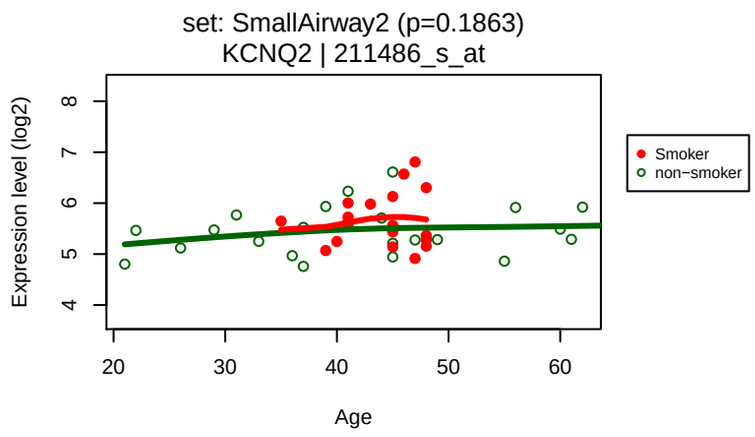
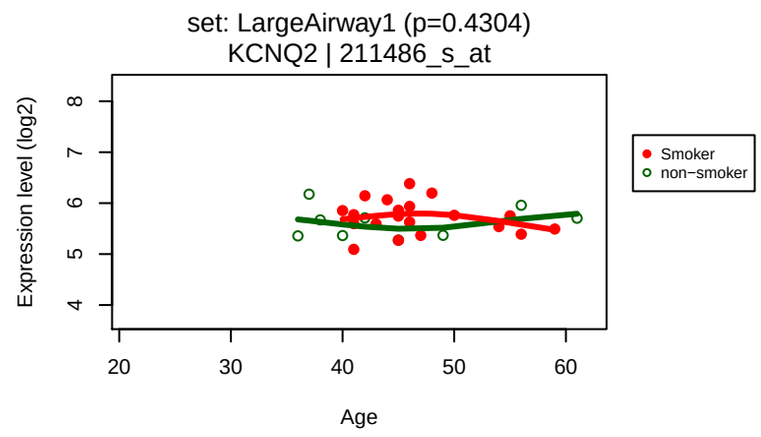

Gene expression of smokers and nonsmokers
gene: KCNQ2 | 211486_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0505
p=0.1863
p=0.1881
p=0.4304
p=0.9053
p=0.0093
p=0.6319
p=0.8703
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

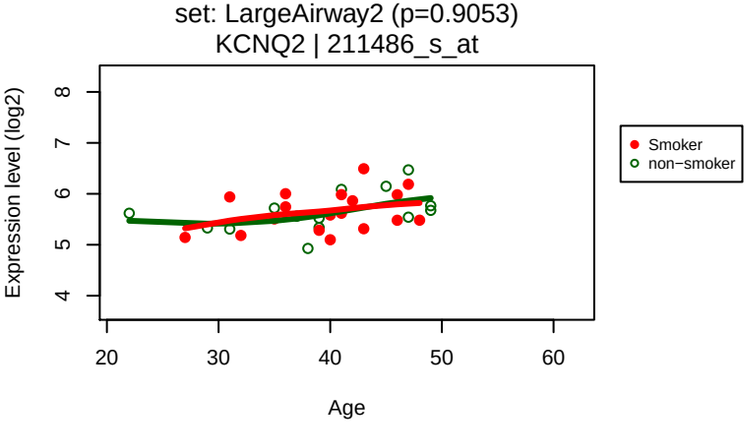

Gene expression ratio (smokers/nonsmokers)
gene: KCNQ2 | 211486_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0505
p=0.1863
p=0.1881
p=0.4304
p=0.9053
p=0.0093
p=0.6319
p=0.8703
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

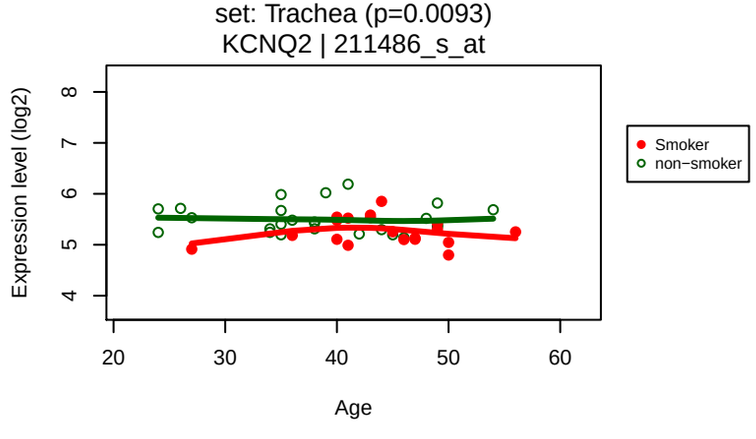

Gene expression ratio in Males
gene: KCNQ2 | 211486_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5351
p=0.4015
p=0.0388
p=0.7741
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KCNQ2 | 211486_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2441
p=0.1509
p=0.2299
p=0.9787
SmallAirway2
LargeAirway2
Trachea
Macrophage2

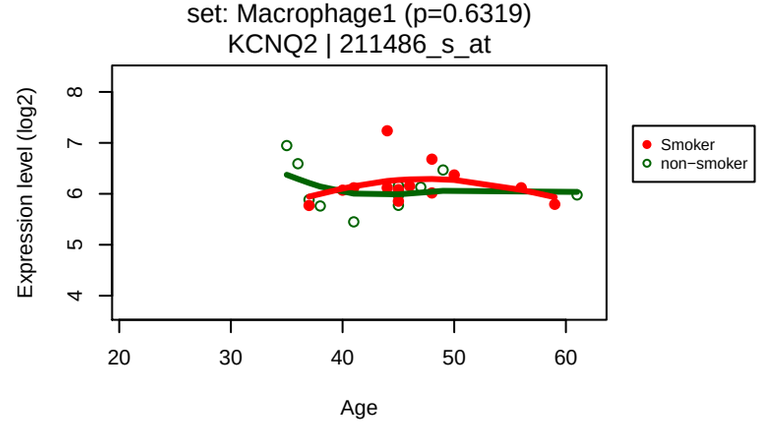

set: SmallAirway1 (p=0.0505)
KCNQ2 | 211486_s_at
Expression level (log2)
Age
Smoker
non−smoker

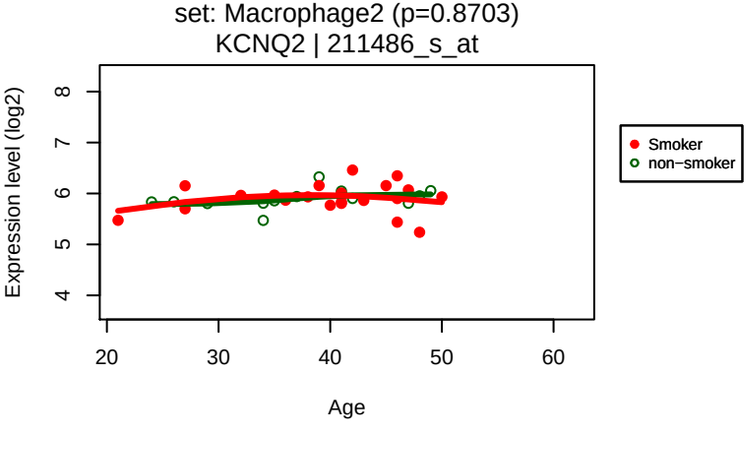

set: SmallAirway2 (p=0.1863)
KCNQ2 | 211486_s_at
Expression level (log2)
Age
Smoker
non−smoker

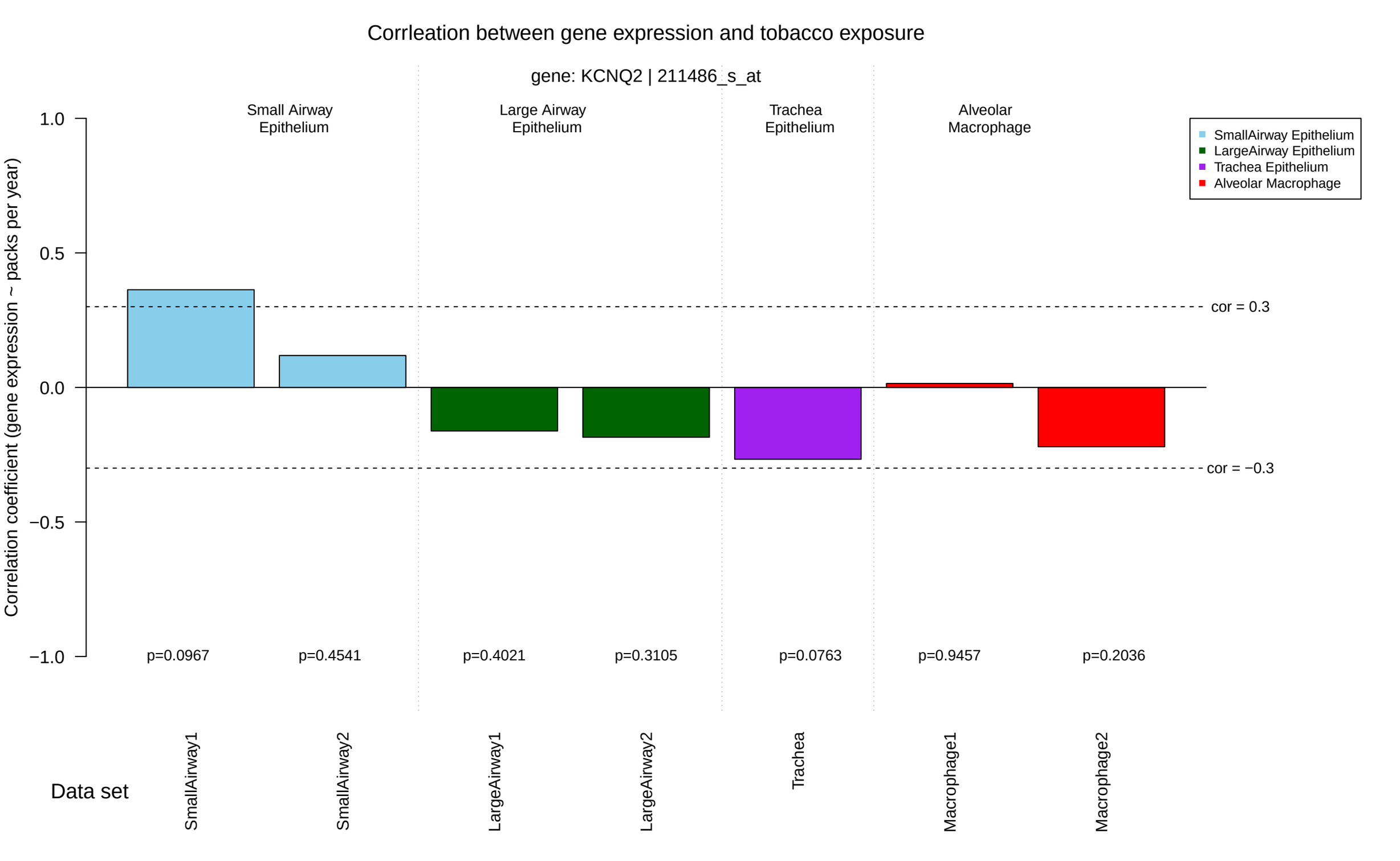

set: LargeAirway1 (p=0.4304)
KCNQ2 | 211486_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9053)
KCNQ2 | 211486_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0093)
KCNQ2 | 211486_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6319)
KCNQ2 | 211486_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8703)
KCNQ2 | 211486_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KCNQ2 | 211486_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0967
p=0.4541
p=0.4021
p=0.3105
p=0.0763
p=0.9457
p=0.2036
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2